MATH60005/70005: Optimisation Coursework (AY 26–27)

Instructions: read this first![1]

- This coursework has a total of 20 marks and accounts for 10% of the module.
- **Submission deadline:** Monday, 23 November 2026, 13:00 UK time, via Blackboard dropbox.
- **Submit a single PDF file typeset in LaTeX, Word, or similar.**
- The coursework requires basic coding capabilities that are available in all modern languages. **You are welcome to use any language of your preference.** You don't need to include your code in the submission; we want to assess your answers/plots/mathematical reasoning.
- **Marking criteria:** Full marks will be awarded for work that 1) is mathematically correct, 2) shows an understanding of material presented in lectures, 3) gives details of all calculations, implementation, and reasoning, and 4) is presented in a logical and clear manner.
- Do not discuss your answers publicly via our forum. If you have any queries regarding your interpretation of the questions, please contact the lecturer at `dkaliseb@imperial.ac.uk`. Consulting with the lecturer excludes debugging code or checking whether an answer is correct.
- Beware of plagiarism regulations; the College takes this very seriously. This is a **group-based assessment** with groups of 1 to 3 students. **Make a single group submission displaying the CID of every group member on the front page. Do not include your name.**

Part I: Robust Regression with Outliers (10 marks)

We want to approximate a function $V : [-1, 1] \to \mathbb{R}$ from $m = 60$ samples $\mathcal{D} = \{x_i, y_i\}_{i=1}^m$ (file `trainingI.dat`[2]), where $y_i \approx V(x_i)$ but a small number of measurements are corrupted by gross errors (*outliers*). We use models

$$V_\theta(x) = \sum_{j=1}^{n} \theta_j P_{j-1}(x) = \theta^\top \Phi(x),$$

where P_k is the Legendre polynomial of degree k, generated by

$$\begin{aligned} P_0 &= 1, \\ P_1 &= x, \\ (k+1)P_{k+1}(x) &= (2k+1)xP_k(x) - kP_{k-1}(x). \end{aligned}$$

Throughout, take $n = 8$, and let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$ be the matrix and vector of the linear least squares problem $\min_\theta \|\mathbf{A}\theta - \mathbf{b}\|_2^2$ associated with this fit. Validation errors are measured as MSE $:= \frac{1}{m_{val}} \sum_{k=1}^{m_{val}} |V_\theta(x_k) - V(x_k)|^2$ on the (clean) data in `validationI.dat`.

[1] The ability to read and follow instructions is a highly valued skill in the job market and elsewhere.

[2] Data format is (x_i, y_i).

I.a) **[3 marks]** Give precise expressions for $\mathbf{A}$ and $\mathbf{b}$, and compute the least squares fit θ_{LS} by solving the normal equations $\mathbf{A}^\top\mathbf{A}\,\theta = \mathbf{A}^\top\mathbf{b}$. Plot the data together with $V_{\theta_{LS}}$ and report its validation MSE. Looking at the residuals $\mathbf{A}\theta_{LS} - \mathbf{b}$, explain why the least squares fit is sensitive to outliers.

Conditioning. For a symmetric positive definite matrix $\mathbf{M}$, the *condition number* is

$$\kappa(\mathbf{M}) := \frac{\lambda_{\max}(\mathbf{M})}{\lambda_{\min}(\mathbf{M})} \;\geq 1.$$

It quantifies how sensitive the solution of a linear system is to perturbations of the data: if $\mathbf{M}\theta = \mathbf{c}$ and $\mathbf{M}(\theta + \delta\theta) = \mathbf{c} + \delta\mathbf{c}$, then

$$\frac{\|\delta\theta\|_2}{\|\theta\|_2} \leq \kappa(\mathbf{M})\,\frac{\|\delta\mathbf{c}\|_2}{\|\mathbf{c}\|_2},$$

so that relative errors in $\mathbf{c}$ (measurement noise, rounding errors) can be amplified by a factor up to $\kappa(\mathbf{M})$ in the solution. Prove this inequality. Compute $\kappa(\mathbf{A}^\top\mathbf{A})$ for the Legendre basis and for the monomial basis $(1, x, \ldots, x^{n-1})$ used in last year's coursework, and comment on which basis you would rather use.

I.b) **[3 marks]** To reduce the influence of outliers, we replace the square loss by the Huber loss

$$\min_{\theta\in\mathbb{R}^n} F_\epsilon(\theta) := \sum_{i=1}^{m} \mathcal{L}_\epsilon\big((\mathbf{A}\theta - \mathbf{b})_i\big), \qquad \mathcal{L}_\epsilon(t) := \begin{cases} \frac{1}{2}t^2 & |t| \leq \epsilon\,, \\ \epsilon\big(|t| - \frac{1}{2}\epsilon\big) & \text{otherwise,} \end{cases} \qquad \epsilon > 0.$$

i) Show that $\mathcal{L}_\epsilon$ is continuously differentiable, and that it is convex. Deduce that F_ϵ is convex and compute $\nabla F_\epsilon(\theta)$ in terms of $\mathbf{A}$, $\mathbf{b}$ and $\mathcal{L}'_\epsilon$.

ii) Show that if $\epsilon \geq \|\mathbf{A}\theta_{LS} - \mathbf{b}\|_\infty$, then θ_{LS} is a global minimiser of F_ϵ. Interpret this result, and the behaviour of $\epsilon^{-1}\mathcal{L}_\epsilon(t)$ as $\epsilon \to 0$.

iii) Is the condition $\epsilon \geq \|\mathbf{A}\theta_{LS} - \mathbf{b}\|_\infty$ also *necessary* for θ_{LS} to minimise F_ϵ? Prove it, or give a counterexample. Then check numerically what happens with our data for ϵ slightly below $\|\mathbf{A}\theta_{LS} - \mathbf{b}\|_\infty$.

I.c) **[4 marks]** Implement the gradient method with constant stepsize $t = 1/\sigma_{\max}(\mathbf{A})^2$, where $\sigma_{\max}(\mathbf{A})$ is the largest singular value of $\mathbf{A}$ (i.e. $\sigma_{\max}(\mathbf{A})^2 = \lambda_{\max}(\mathbf{A}^\top\mathbf{A})$), to minimise F_ϵ. Describe your initialisation and stopping criterion.

i) For $\epsilon = 10^{-3}, 10^{-2.5}, \ldots, 10^{1}$, report the number of iterations and plot the validation MSE versus ϵ (log-log scale), indicating the least squares MSE for reference. Show the fit for your best ϵ, and use it to identify the outliers in `trainingI.dat` (report their row numbers). What do you observe for small and for large ϵ?

ii) The number of iterations grows as ϵ decreases. Let θ^* be a minimiser of F_ϵ such that no residual $(\mathbf{A}\theta^* - \mathbf{b})_i$ equals $\pm\epsilon$, and let $\mathcal{I} := \{i : \; |(\mathbf{A}\theta^* - \mathbf{b})_i| < \epsilon\}$. Show that in a neighbourhood of θ^*, F_ϵ is a quadratic function with Hessian $\mathbf{A}_\mathcal{I}^\top\mathbf{A}_\mathcal{I}$, where $\mathbf{A}_\mathcal{I}$ collects the rows of $\mathbf{A}$ indexed by $\mathcal{I}$. Compute $|\mathcal{I}|$ and $\lambda_{\min}(\mathbf{A}_\mathcal{I}^\top\mathbf{A}_\mathcal{I})$ for each ϵ, plot $F_\epsilon(\theta^k) - F_\epsilon(\theta^*)$ versus k (semi-log scale) for $\epsilon \in \{10^{-2}, 10^{-1}, 1\}$, and use these results to explain the iteration counts.

Part II: Tracking a Moving Target from Range Measurements (10 marks)

In the exercise class we estimated a *static* position from range measurements by linearising the range equations. Here we track a *moving* target, and we go beyond the linearisation. Three anchors at known positions $\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3 \in \mathbb{R}^2$ lie along a nearly straight coastline (file `anchorsII.dat`,

one anchor per row). A target moves along an unknown trajectory $\mathbf{p}_1, \dots, \mathbf{p}_T \in \mathbb{R}^2$, $T = 60$. At each time t, anchor i measures a noisy distance $r_{t,i} \approx \|\mathbf{p}_t - \mathbf{s}_i\|_2$ (file `rangesII.dat`, a $T \times 3$ array whose row t is $(r_{t,1}, r_{t,2}, r_{t,3})$). The true position is known only at six checkpoints (file `checkpointsII.dat`, format $(t, p_{t,1}, p_{t,2})$). Use the checkpoints only for validation, through the error

$$E_{ck} := \Big(\tfrac{1}{6} \sum_{t \in \{10,20,\dots,60\}} \|\widehat{\mathbf{p}}_t - \mathbf{p}_t\|_2^2 \Big)^{1/2}.$$

At each time t we consider the *squared-range least squares* function

$$g_t(\mathbf{p}) := \frac{1}{4} \sum_{i=1}^{3} \big(\|\mathbf{p} - \mathbf{s}_i\|_2^2 - r_{t,i}^2 \big)^2,$$

and the linearised estimate from the exercise class, $\mathbf{p}_{LLS,t} := \arg\min_{\mathbf{p}} \|\mathbf{C}\mathbf{p} - \mathbf{d}_t\|_2^2$, where $\mathbf{C}$ has rows $2(\mathbf{s}_i - \mathbf{s}_1)^\top$ and $\mathbf{d}_t$ has entries $r_{t,1}^2 - r_{t,i}^2 + \|\mathbf{s}_i\|_2^2 - \|\mathbf{s}_1\|_2^2$, for $i = 2, 3$.

II.a) **[2 marks]** Consider first a single anchor at the origin, $g(\mathbf{p}) = \frac{1}{4}(\|\mathbf{p}\|_2^2 - r^2)^2$ with $r > 0$. Find all the stationary points of g and classify them according to whether they are saddle points, strict/nonstrict local/global minimum/maximum points. Is g convex? Show that each g_t is coercive, and conclude that it has a global minimiser.

II.b) **[2 marks]** You may use without proof that

$$\nabla g_t(\mathbf{p}) = \sum_{i=1}^{3} \big(\|\mathbf{p}-\mathbf{s}_i\|_2^2 - r_{t,i}^2 \big)(\mathbf{p}-\mathbf{s}_i), \qquad \nabla^2 g_t(\mathbf{p}) = \sum_{i=1}^{3} \Big[2(\mathbf{p}-\mathbf{s}_i)(\mathbf{p}-\mathbf{s}_i)^\top + \big(\|\mathbf{p}-\mathbf{s}_i\|_2^2 - r_{t,i}^2 \big)\mathbf{I} \Big].$$

i) Show that $\nabla^2 g_t(\mathbf{p}) \succeq 0$ on $\Omega_t := \{\mathbf{p} : \ \|\mathbf{p} - \mathbf{s}_i\|_2 \geq r_{t,i}, \ i = 1, 2, 3\}$. Is g_1 convex on $\mathbb{R}^2$? Justify your answer by evaluating $\nabla^2 g_1$ at the three anchors.

ii) Suppose instead that all anchors lay exactly on the horizontal axis. Show that $g_t(p_1, -p_2) = g_t(p_1, p_2)$. Deduce that the mirror image of a stationary point is also a stationary point, of the same type. What do you expect for our nearly collinear anchors?

II.c) **[2 marks]** *Static estimation.* For each t, compute $\mathbf{p}_{LLS,t}$. Then run the gradient method with constant stepsize 10^{-3} on g_t, initialised at $\mathbf{p}_{LLS,t}$, and state your stopping criterion. Plot both estimated trajectories together with the anchors and the checkpoints, and report E_{ck} for each. Compute $\kappa(\mathbf{C}^\top\mathbf{C})$ (see Part I.a). Explain the accuracy of $\mathbf{p}_{LLS,t}$, and why some of the gradient method estimates jump between the two sides of the coastline. Identify the time instants where this happens.

II.d) **[4 marks]** *Tracking.* To exploit the fact that the target moves continuously, we estimate the whole trajectory $\mathbf{P} \in \mathbb{R}^{T \times 2}$ (row t is $\mathbf{p}_t^\top$) by solving

$$\min_{\mathbf{P} \in \mathbb{R}^{T \times 2}} \ G_\lambda(\mathbf{P}) := \sum_{t=1}^{T} g_t(\mathbf{p}_t) + \frac{\lambda}{2} \sum_{t=1}^{T-1} \|\mathbf{p}_{t+1} - \mathbf{p}_t\|_2^2, \qquad \lambda \geq 0.$$

i) Write the regularisation term as $\frac{\lambda}{2}\|\mathbf{D}\mathbf{P}\|_F^2$, giving $\mathbf{D} \in \mathbb{R}^{(T-1)\times T}$ explicitly. Show that it is a convex function of $\mathbf{P}$, and compute $\nabla G_\lambda(\mathbf{P})$ in matrix form. Is G_λ convex?

ii) Run the gradient method with constant stepsize $1/(1000 + 4\lambda)$, initialised at your static estimates from II.c), for $\lambda \in \{10^{-1}, 1, 10, 10^2, 10^3\}$. Plot E_{ck} versus λ, select λ, and plot the resulting trajectory. Discuss under- and over-regularisation. What do you expect the minimisers of G_λ to look like as $\lambda \to \infty$?

iii) For your selected λ, let $\mathbf{P}^*$ be the trajectory obtained in ii), and run the method again initialised at its mirror image, i.e. with each $\mathbf{p}_t^* = (p_{t,1}^*, p_{t,2}^*)$ replaced by $(p_{t,1}^*, -p_{t,2}^*)$ as in II.b ii). Compare the two limits in terms of G_λ, E_{ck} and the trajectories, and explain the result using II.b). Would the regularisation be enough to determine the trajectory if the anchors were exactly collinear? What information resolves the ambiguity?